

```
1  --B44 internal
2  --PS-5 : range and hash partitioning
3
4  CREATE TABLE Movie (
5      movie_id      NUMBER PRIMARY KEY,
6      title         VARCHAR2(100),
7      genre         VARCHAR2(50),
8      language      VARCHAR2(30),
9      duration      NUMBER,
10     rating         VARCHAR2(10),
11     release_year   NUMBER
12 );
13
14
15 CREATE TABLE Customer (
16     customer_id   NUMBER PRIMARY KEY,
17     name          VARCHAR2(100),
18     email         VARCHAR2(100),
19     phone         VARCHAR2(15),
20     city          VARCHAR2(50)
21 );
22
23
24 CREATE TABLE Theatre (
25     theatre_id    NUMBER PRIMARY KEY,
26     theatre_name  VARCHAR2(100),
27     city          VARCHAR2(50),
28     screens       NUMBER
29 );
30
31
32 CREATE TABLE Seat (
33     seat_id       NUMBER PRIMARY KEY,
34     seat_number   VARCHAR2(10),
35     seat_type     VARCHAR2(20)
36 );
37
38
39 CREATE TABLE Show (
40     show_id       NUMBER PRIMARY KEY,
41     movie_id      NUMBER REFERENCES Movie(movie_id),
42     theatre_id    NUMBER REFERENCES Theatre(theatre_id),
43     show_date     DATE,
44     show_time     VARCHAR2(10),
45     ticket_price  NUMBER
46 );
47
48
49 CREATE TABLE Booking (
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50      booking_id      NUMBER PRIMARY KEY,
51      customer_id     NUMBER REFERENCES Customer(customer_id),
52      show_id         NUMBER REFERENCES Show(show_id),
53      seat_id         NUMBER REFERENCES Seat(seat_id),
54      booking_date    DATE,
55      total_amount    NUMBER,
56      payment_status  VARCHAR2(20)
57 );
58
59
60 CREATE TABLE Booking_Range (
61     booking_id      NUMBER PRIMARY KEY,
62     customer_id     NUMBER,
63     show_id         NUMBER,
64     seat_id         NUMBER,
65     booking_date    DATE,
66     total_amount    NUMBER,
67     payment_status  VARCHAR2(20)
68 )
69 PARTITION BY RANGE (booking_date) (
70     PARTITION bk_2023 VALUES LESS THAN (DATE '2024-01-01'),
71     PARTITION bk_2024 VALUES LESS THAN (DATE '2025-01-01'),
72     PARTITION bk_2025 VALUES LESS THAN (DATE '2026-01-01'),
73     PARTITION bk_future VALUES LESS THAN (MAXVALUE)
74 );
75
76
77
78
79 CREATE TABLE Customer_Hash (
80     customer_id     NUMBER PRIMARY KEY,
81     name            VARCHAR2(100),
82     email           VARCHAR2(100),
83     phone           VARCHAR2(15),
84     city            VARCHAR2(50)
85 )
86 PARTITION BY HASH (customer_id) (
87     PARTITION cust_p1,
88     PARTITION cust_p2,
89     PARTITION cust_p3,
90     PARTITION cust_p4
91 );
92
93
94 INSERT INTO Movie VALUES (1, 'RRR', 'Action', 'Telugu', 187, 'U/A', 2022);
95 INSERT INTO Movie VALUES (2, 'KGF', 'Action', 'Kannada', 139, 'U/A', 2018);
96
97
98 INSERT INTO Theatre VALUES (1, 'PVR Cinemas', 'Kolhapur', 5);
99 INSERT INTO Theatre VALUES (2, 'INOX', 'Pune', 8);
100
```

```
101
102 INSERT INTO Customer VALUES (1, 'Rahul Sharma', 'rahul@email.com',
103                                '9876543210', 'Kolhapur');
104 INSERT INTO Customer VALUES (2, 'Priya Patil', 'priya@email.com',
105                                '9123456780', 'Pune');
106 INSERT INTO Customer VALUES (3, 'Amit Desai', 'amit@email.com',
107                                '9988776655', 'Kolhapur');
108
109 INSERT INTO Seat VALUES (1, 'A1', 'Regular');
110 INSERT INTO Seat VALUES (2, 'A2', 'Regular');
111 INSERT INTO Seat VALUES (3, 'B1', 'Premium');
112
113 INSERT INTO Show VALUES (1, 1, 1, DATE '2025-03-10', '10:00 AM', 200);
114 INSERT INTO Show VALUES (2, 1, 2, DATE '2025-03-10', '01:00 PM', 250);
115 INSERT INTO Show VALUES (3, 2, 1, DATE '2025-03-11', '06:00 PM', 300);
116
117 INSERT INTO Booking VALUES (1, 1, 1, 1, DATE '2025-03-09', 200, 'PAID');
118 INSERT INTO Booking VALUES (2, 2, 2, 2, DATE '2025-03-09', 250, 'PAID');
119 INSERT INTO Booking VALUES (3, 3, 3, 3, DATE '2025-03-10', 300, 'PENDING');
120
121 COMMIT;
122
123 INSERT INTO Booking_Range VALUES (1, 101, 201, 301, DATE '2023-06-15', 500,
124                                    'PAID');
125 INSERT INTO Booking_Range VALUES (2, 102, 202, 302, DATE '2024-03-20', 300,
126                                    'PAID');
127 INSERT INTO Booking_Range VALUES (3, 103, 203, 303, DATE '2025-01-10', 450,
128                                    'PENDING');
129 COMMIT;
130
131 INSERT INTO Customer_Hash VALUES (1, 'Rahul Sharma', 'rahul@email.com',
132                                    '9876543210', 'Kolhapur');
133 INSERT INTO Customer_Hash VALUES (2, 'Priya Patil', 'priya@email.com',
134                                    '9123456780', 'Pune');
135 INSERT INTO Customer_Hash VALUES (3, 'Amit Desai', 'amit@email.com',
136                                    '9988776655', 'Mumbai');
137 INSERT INTO Customer_Hash VALUES (4, 'Sneha More', 'sneha@email.com',
138                                    '9001234567', 'Nashik');
139 COMMIT;
140
141 SELECT * FROM Booking_Range PARTITION (bk_2023);
142 SELECT * FROM Booking_Range PARTITION (bk_2024);
143 SELECT * FROM Booking_Range PARTITION (bk_2025);
144
145 SELECT * FROM Customer_Hash PARTITION (cust_p1);
146 SELECT * FROM Customer_Hash PARTITION (cust_p2);
147 SELECT * FROM Customer_Hash PARTITION (cust_p3);
148 SELECT * FROM Customer_Hash PARTITION (cust_p4);
```

```
142
143
144
145 SELECT * FROM Show WHERE movie_id = 1;
146
147
148 SELECT * FROM Booking WHERE payment_status = 'PAID';
149
150
151 SELECT * FROM Booking WHERE customer_id = 1;
152
153
154 SELECT COUNT(*) AS total_bookings FROM Booking;
155
156
157 SELECT * FROM Theatre WHERE city = 'Kolhapur';
```